

Conservation Genetics Quiz #4

Name: _____

- 1) Draw simple graphs depicting the relationship between F_{ST} and a) the ratio of observed to expected average heterozygosity of subpopulations; b) migration rate between subpopulations when population size is constant; c) effective population size of subpopulations when migration is constant (3 points).

- 2) Explain the One-Migrant-Per-Generation Rule using the equation $F_{ST} = \frac{1}{4Nm+1}$. Hint: You may wish to apply it to two populations of different effective sizes with different migration rates. (4 points)

- 3) What does the 50 in Franklin's 50/500 rule refer to? What variables were considered in its derivation? (3 points)

- 4) What does the 500 in Franklin's 50/500 rule refer to? What variables were considered in its derivation? (3 points)

- 5) Mountain Caribou are a subspecies of Woodland Caribou found in inland temperate rainforest in north-eastern Washington, northern Idaho, and British Columbia. In 2009, the South Selkirks subpopulation of Mountain Caribou numbered 46 individuals.
- a. If observed heterozygosity in this population was 0.2 in 2009 and 0.18 in 2021, what is the species' N_e/N ratio? (Caribou have a generation time of 6 years; the equation $H_t = H_o e^{\frac{-t}{2N_e}}$ may be useful.) (5 points)
- b. At the time the heterozygosity was measured in 2021, allele frequencies at a diallelic locus of interest were $f(A_1) = 0.8$ and $f(A_2) = 0.2$. What is the inbreeding coefficient for this locus? Remember that F can be calculated as $1 - \frac{H_o}{H_e}$. (2 points)
- 5) Hermit Warblers and Townsend's Warblers are small songbirds with locally overlapping ranges in the mountains of the Pacific Northwest, where they hybridize. Assume that you are a graduate student studying the species using diagnostic nuclear markers, such that pure Hermit Warblers have the genotypes $AA/BB/CC/DD$, and pure Townsend's Warblers have the genotypes $aa/bb/cc/dd$.
- a. A phenotypically ambiguous warbler is genotyped as $Aa/bb/CC/Dd$. Remember that hybrids can be classified as F1 individuals, F2 individuals, or late-generation hybrids (F3 and on). Your advisor claims the bird must be an F1 (e.g., the product of a cross between pure Hermit and pure Townsend's individuals). Do you agree? Why or why not? (3 points)
- b. What is the probability an F2 hybrid is homozygous at all four diagnostic loci? (2 points)